	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering Sector No. 26, Pradhikaran, Nigdi, Pune—411 044
Question Bank	

Department: **Computer Engineering** AY: **2025-26** Semester: **II** Class : **SY B Tech**
Course Name & Code: **Advanced Data Structures**
Course Owner and Faculties: Mrs Swati Chandurkar (CO), Mrs Ashwini Matange , Dr. Reena Kharat and Dr Aparna Joshi

Important Instructions for students:

- Below question bank contains sample questions.
- Not mandatory that same questions will reflect in SA exam.
- Students are expected to prepare accordingly.

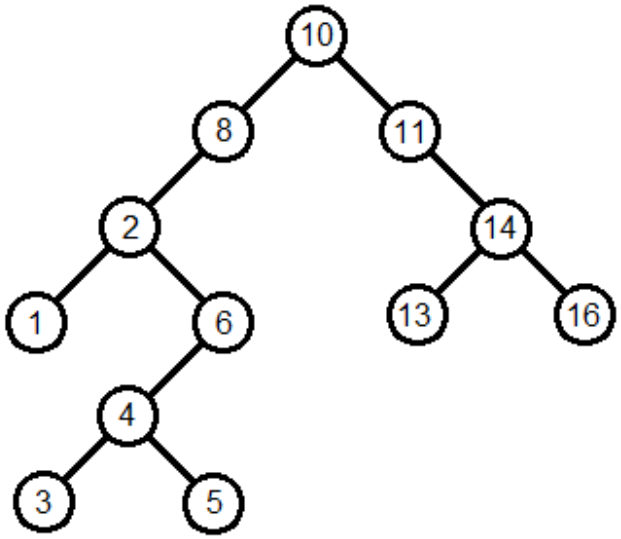
Course Outcomes:
After learning the course, the students will be able to:
1. Apply the concepts of Binary Trees for problem solving
2. Apply the concepts of advanced binary trees for various problem solving
3. Apply the concepts of Graphs for solving real life problems
4. Understand the concepts of advanced search trees.

Question Bank Summary				
	Count of Easy Questions	Count of Moderate Questions	Count of Difficult Questions	Total no of Questions
Unit 1	**9**	**13**	**8**	**30**
Unit 2	9	8	5	22
Unit 3	7	20	6	33
Unit 4	10	9	5	24
Total	35	50	24	109

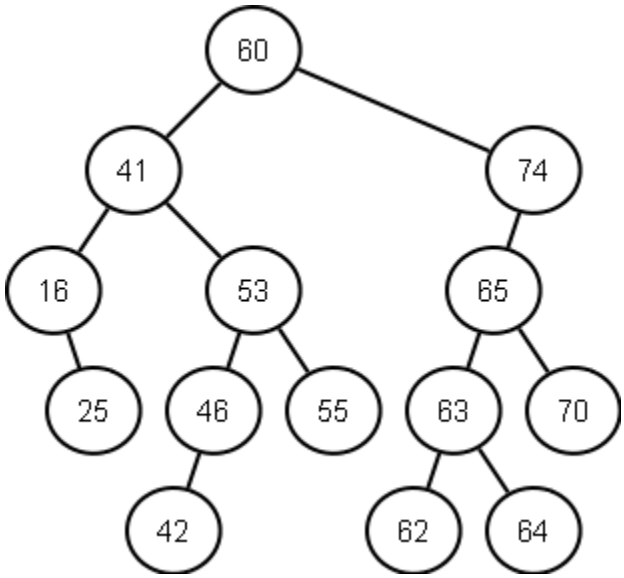
Unit 1

UNIT 1: I*ntroduction to Tree: Basic terminologies, representation using array and linked list, Introduction to Binary Tree; Properties of Binary Tree; Operations on Binary tree: Finding Height, Finding Leaf nodes, counting no of Nodes; Recursive and Nonrecursive traversals; Construction of Binary Tree from traversals; Binary Search tree (BST) as an ADT: Insertion, deletion and searching of a node from BST.*

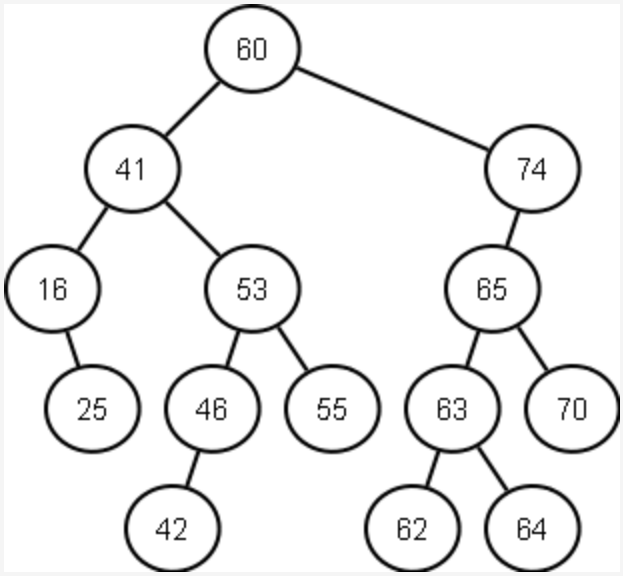
BL3 verbs: **act administer apply articulate calculate change chart choose collect complete compute construct determine develop discover dramatize employ establish examine experiment explain illustrate interpret judge manipulate modify operate practice predict prepare produce record relate report schedule simulate sketch solve teach transfer write**

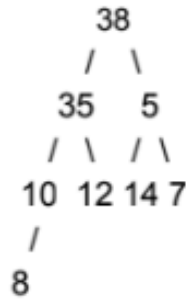
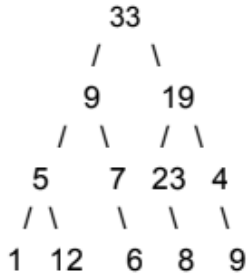
S No	Questions	Marks	Difficulty level	CO Mapping	GATE	BL
1	A binary tree is stored in an array starting at index 1. If a node is at index *i*, determine the indices of its left child, right child, and parent. Illustrate this with a tree of height 2.	2	**Easy**	**CO1**		BL 3
2	 For the above BST, trace the comparisons made to search for element 5 and 12	2	**Easy**	**CO1**		BL 3
3	A binary tree has **In-order:** D B E A F C and **Pre-order:** A B D E C F. Construct the tree and explain the logic used to find the root.	3	**Easy**	**CO1**		BL 3

4	//preorder traversal
	20, 15, -5, -150, 0, 12, 200, 25, 20, 100, 126, 1000
	
	// inorder traversal
	-150, -5, 0, 12, 15, 20, 20, 25, 100, 126, 200, 1000
	
	From the above two traversals, draw the binary tree.
		3	**Easy**	**CO1**		BL 3

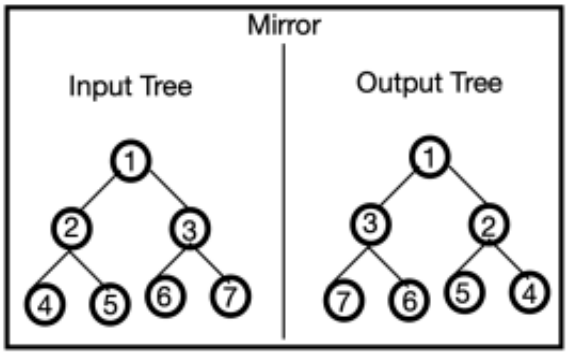
5	Insert the sequence [45, 20, 70, 10, 30, 60, 85] into an initially empty BST. Draw the resulting tree.	3	**Easy**	**CO1**		BL 3
6	Insert the elements 71,65,84,69,67,83 in an empty binary search tree(BST)in the sequence shown. How many levels does the tree have? Which element is at the lowest level?	3	**Easy**	**CO1**	Gate	BL 3
7	The preorder traversal sequence of a binary search tree is 30,20,10 ,15, 25, 23, 39, 35,42. Which one of the following is the postorder traversal sequence of the same tree? A] 10, 20, 15, 23, 25, 35, 42, 39, 30 B] 15, 10, 25, 23, 20, 42, 35, 39, 30 C] 15, 20, 10, 23, 25, 42, 35, 39, 30 D] 15, 10, 23, 25, 20, 35, 42, 39, 30	2	**Easy**	**CO1**	GATE	BL 3
8		3	**Easy**	**CO1**		BL 3

	Redraw the tree after deleting the root. Explain logic applied					
9	The inorder and preorder traversal of a binary tree are ***d b e a f c g*** and ***a b d e c f g*** respectively. Determine the postorder traversal of the tree.		**Easy**	**CO1**	Gate	BL 3
10	Write a function **checkMirrorImage()**, which will take the root nodes of the given binary trees and return True/False depending on whether the trees are mirror images or not of each other. The input nodes for the binary tree will be given in level order (filled up from the leftmost node). **Sample Input:** Number of elements: 7 Enter the elements of Tree 1: 10 15 32 7 9 11 4 Enter the elements of Tree 2: 10 32 15 4 11 9 7 **Sample Output:** Yes, Tree 1 and Tree 2 are mirror images of each other. Sample trees for reference: Tree 1 Mirror Tree 2 10 | 10 / \ | / \ 15 32 | 32 15 / \ / \ | / \ / \ 7 9 11 4 | 4 11 9 7	4	**Difficult**	**CO1**		BL 3
11	Write a function to display all nodes from root to the farthest leaf. For example, if the input tree is as below, the desired output is: 60 41 53 46 42 **OR** 60 74 85 63 62 **OR** 60 74 85 63 64.	4	**Difficult**	**CO1**		BL 3

						
12	Write a function to find the least/lowest common ancestor of two nodes in a complete binary tree. You can take the input in level order from left to right. All the tree entries will be unique. NOTE: The lowest/least common ancestor is defined between two nodes p and q as the lowest node in the Tree that has both p and q as descendants. Expected Input: Enter number of elements: 8 Enter the elements: 38 35 5 10 12 14 7 8 Element 1: 10 Element 2: 8 Expected Output: The LCA of 10 and 8 is 35 Element 1: 12 Element 2: 14 Expected Output: The LCA of 12 and 14 is 38 For your reference: The tree looks like this:	6	**Difficult**	**CO1**		BL 3

						
13	Write a function to display all nodes of a subtree rooted at node ***x.***	4	**Difficult**	**CO1**		BL 3
14	Write a function for a **Non-recursive Pre-order** traversal of a Binary Tree. Assume all supporting Data Structures are available.	6	**Difficult**	**CO1**		BL 3
15	 Write a function to determine whether the tree has a path from the root to the leaf of a given sum. //For the above tree Sample Input: 55 Sample Output: Yes [For your reference, the path is 33-9-7-6] Sample Input: 35 Sample Output : No	6	**Difficult**	**CO1**		BL 3
16	Write a function to check if the given input binary tree is a BST.	4	**Difficult**	**CO1**		BL 3
17	Write a deleteNode(Node* root, int key) function, handling all possible cases.	6	**Difficult**	**CO1**		BL 3

18	In a perfect binary tree of height ***h***, calculate the total number of nodes and the number of leaf nodes. Show the steps for ***h = 4***	4	**Moderate**	**CO1**		BL 3
19	If a binary tree has 15 nodes, calculate the minimum and maximum possible heights it can have.	4	**Moderate**	**CO1**		BL 3
20	Write a function **int countLeaf(Node* root)** that uses recursion to return the number of leaf nodes in a binary tree.	4	**Moderate**	**CO1**		BL 3
21	Compare the Array representation and Linked List representation of a binary tree in terms of memory wastage for a skewed tree. Provide a calculation example for a tree of height 5.	4	**Moderate**	**CO1**		BL 3
22	Show how the expression (a + b) * (c - d) would be represented as a binary tree. List its Post-order traversal.	4	**Moderate**	**CO1**		BL 3
23	Explain how a **Queue** is used to perform a Level Order Traversal. Trace it for a tree with 3 levels.	**4**	**Moderate**	**CO1**		BL 3
24	Write a logic to determine if a given binary tree is a **Complete Binary Tree** using its array indices.	**4**	**Moderate**	**CO1**		BL 3
25	Write a function to count the total number of non-leaf (internal) nodes in a binary tree.	**4**	**Moderate**	**CO1**		BL 3
26	Given the infix expression: (29 – 3) * 4 / 6 + 68 % (3 + 10) Create an expression tree that represents this expression. The expression tree is a binary tree where all leaf nodes are operands, and all non-leaf nodes are operators. () : precedence 1, associativity left to right /, *, % : precedence 2, associativity left to right +, - : precedence 3, associativity left to right Show the postfix expression represented by the tree, with spaces between each token and no leading or trailing spaces.	**4**	**Moderate**	**CO1**		BL 3
27	Complete the implementation of a function that converts a given binary tree to its mirror, as shown in the figure below. Fill out the blank spaces. Node definition is given for your reference.	**4**	**Moderate**	**CO1**		BL 3

	struct Node {
	int data;
	Node* left;
	Node* right;
	};
	void mirror(struct Node* root) {
	if (root==NULL) return;
	else {
	Node* temp;
	mirror (___); // Fill out the blank space
	mirror (___); // Fill out the blank space
	temp = ____; // Fill out the blank space
	root->_____ = root->_____; // Fill out the blank spaces
	root->_____ = _____; // Fill out the blank spaces
	}
	}
	Solution sketch:
	 					

28	Write a function to calculate the sum of all the elements in the Binary Tree.	**4**	**Moderate**	**CO1**		BL 3
29	The height of a tree is the length of the longest root-to-leaf path in it. Calculate and explain the maximum and minimum number of nodes in a binary tree of height 5.	**4**	**Moderate**	**CO1**	Gate	BL 3
30	Write a function to retrieve sibling of a node assuming linked representation and node structure as below: struct node{ int key; Node *left, *right;	**4**	**Moderate**	**CO1**		BL 3

	}					

Unit 2

S No	Questions	Marks	Difficulty level	CO Mapping	GATE	BL
1	What is the maximum height of any AVL-tree with 7 nodes? Assume that the height of a tree with a single node is 0. Justify.	05	Easy	CO2		BL 3
2	Write an algorithm for LL rotation in AVL tree. Discuss with example.	05	Easy	CO2		BL 3
3	Write an algorithm for RR rotation in AVL tree.Discuss with example.	05	Easy	CO2		BL 3
4	Write an algorithm for LR rotation in AVL tree.Discuss with example.	05	Easy	CO2		BL 3
5	Write an algorithm for RL rotation in AVL tree.Discuss with example.	05	Easy	CO2		BL 3
6	Discuss heapify algorithm with any example tree.	05	Easy	CO2		BL 3
7	What is the number of rotations required to insert a sequence of elements 9, 6, 5, 8, 7, 10 into an empty AVL tree?	05	Moderate	CO2		BL 3
8	What are Threaded Binary Trees? List its advantages and applications.	05	Easy	CO2		BL 3
9	Why are AVL trees preferred over BST in certain cases?	05	Easy	CO2		BL 3
10	What is a height-balanced tree?	05	Easy	CO2		BL 3
11	Define balance factor in AVL trees. Illustrate with example.	05	Easy	CO2		BL 3

12	Define predecessor and successor in the context of TBT. illustrate with example.	05	Easy	CO2		BL 3
13	Discuss the need for Threaded Binary Trees over conventional binary trees.	05	Easy	CO2		BL 3
14	Compare AVL Tree and Binary Search Tree with example.	04	Moderate	CO2		BL 3
15	Consider a max heap, represented by the array: 40, 30, 20, 10,15,16,17,8,4. New element 35 is inserted in this heap. What would be the array contents after this insertion.	06	Moderate	CO2		BL 3
16	Insert the following nodes in a two way threaded tree. Show all steps. Nodes - 16 10 13 17 20 5	06	Moderate	CO2		BL 3
17	Create a BST and AVL Trees for numbers from 1 to 10. What is the difference in the height of two trees?	06	Moderate	CO2		BL 3
18	How is the Balance Factor calculated? Draw a tree with the numbers 45, 89, 62, 39, 44, 70, Write Balance Factor of each node.	06	Moderate	CO2		BL 3
19	Write an algorithm for inorder traversal of a Threaded Binary Tree.	06	Moderate	CO2		BL 3
20	Write an algorithm for preorder traversal of a Threaded Binary Tree.	06	Moderate	CO2		BL 3
21	Write an algorithm for postorder traversal of a Threaded Binary Tree.	06	Moderate	CO2		BL 3
22	Construct a Max Heap from the following elements: 10, 20, 5, 6, 1, 8, 9.	07	Difficult	CO2		BL 3
23	Insert the following elements into an AVL Tree: 10, 20, 30, 40, 50, 25 and show rotations.	07	Difficult	CO2		BL 3
24	Demonstrate deletion in AVL tree with a suitable example.	07	Difficult	CO2		BL 3
25	Discuss Max Heap and Min Heap with construction, insertion, and deletion	06	Difficult	CO2		BL 3
26	Explain creation and traversal of Threaded Binary Tree with	06	Difficu	CO2		BL

	diagrams.		lt			3
27	Compare AVL Tree with BST.	03	Easy	CO2		BL 2
28	GATE 2006 : The minimum number of nodes in an AVL tree of height h satisfies: Options: A) 2h−12^h - 12h−1 B) 2h+1−12^{h+1} - 12h+1−1 C) N(h−1)+N(h−2)+1N(h-1) + N(h-2) + 1N(h−1)+N(h−2)+1 D) h2h^2h2 ✅ Answer: C	01	Difficult	CO2		BL 3
29	GATE 2018 : Which of the following are height-balanced trees? Options include: BST AVL Red-Black Tree ✅ Answer: AVL and Red-Black	01	Easy	CO2		BL 2
30	GATE 2008 : In an inorder threaded binary tree, a NULL right pointer of a node is replaced by: Options: A) Left child B) Right child C) Inorder successor D) Root ✅ Answer: C) Inorder successor	01	Easy	CO2		BL 2
31	GATE 2012 : Which data structure allows inorder traversal without recursion and without stack? Options: A) Binary Heap B) AVL Tree C) Threaded Binary Tree D) B-Tree ✅ Answer: C) Threaded Binary Tree	01	Easy	CO2		BL 2
32	GATE 2015 : Threaded binary trees are useful because: Options: A) Reduce height	01	Easy	CO2		BL 2

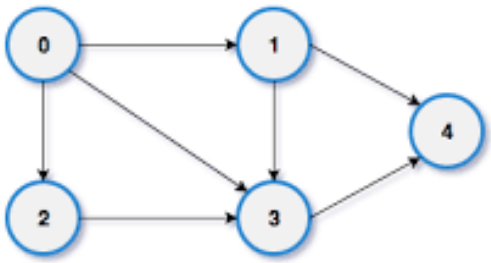
	B) Reduce memory usage C) Utilize NULL pointers D) Improve search time ✅ Answer: C) Utilize NULL pointers					
33	GATE 2007 : A binary heap is: Options: A) Complete binary tree B) Full binary tree C) Perfect binary tree D) Skewed tree ✅ Answer: A) Complete binary tree	01	Easy	CO2		BL 2
34	GATE 2015 : Time complexity to build a heap from an unsorted array: Options: A) O(n log n) B) O(n) C) O(log n) D) O(n^2) ✅ Answer: B) O(n)	01	Easy	CO2		BL 2
35	GATE 2004 : Which of the following is always true for a max heap? Options: A) Root is smallest B) Root is largest C) Left child > right child D) Tree is balanced ✅ Answer: B) Root is largest	01	Easy	CO2		BL 2

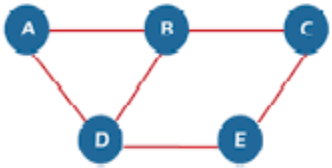
BL3 verbs: **act, administer, apply, articulate, calculate, change, chart, choose, collect, complete, compute, construct, determine, develop, discover, dramatize, employ, establish, examine, experiment, explain, illustrate, interpret, judge, manipulate, modify, operate, practice, predict, prepare, produce, record, relate, report, schedule, simulate, sketch, solve, teach, transfer, write**

Unit 3

S No	Questions	Marks	Difficulty level	CO Mapping	GATE	BL
1	Illustrate the graph representation techniques with example	4	Easy	CO3		BL3
2	Write pseudo code of Dijkstra's algorithm and explain with an example.	3	Easy	CO3		BL3

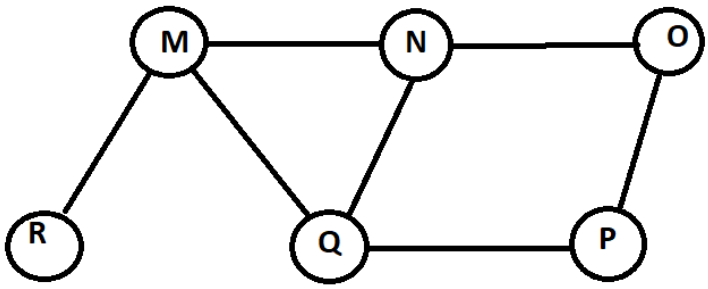
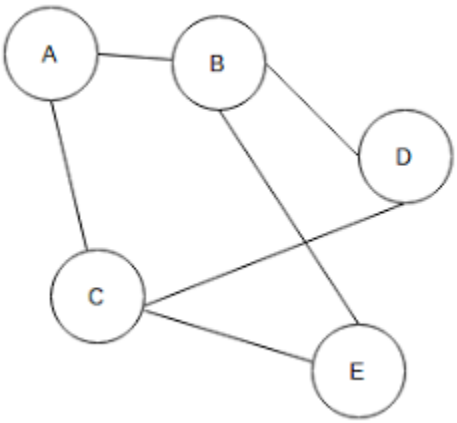
3	Complete the missing statement in the following Prim's Algorithm code:
	`if (visited[i])`
	`{`
	`for (int j = 0; j < V; j++)`
	`{`
	`if (!visited[j] && graph[i][j])`
	`{`
	`if (min > _____)`
	`{`
	`min = graph[i][j];`
	`x = i;`
	`y = j;`
	`}`
	`}`
	`}`
	`}`
		2	Easy	CO3		BL3

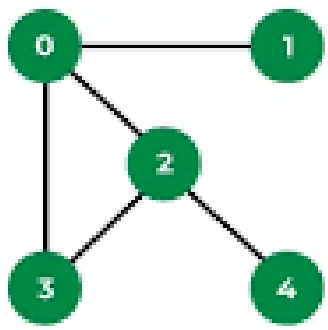
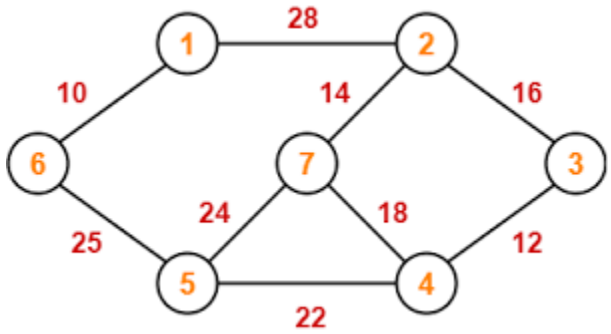
4	Write pseudo code of depth first search traversal and explain with an example.	3	Easy	CO3		BL3
5	Given the following graph:  a) Construct the graph using an adjacency matrix. b) Construct the graph using an adjacency list. c) Suggest the better representation for this graph and justify.	5	Easy	CO3		BL3

6	Given the following network:  a) Apply BFS starting from node A. b) Show level-wise traversal using the diagram.	4	Easy	CO3		BL3
7	Differentiate: ● Weighted graph and unweighted graph ● breadth-first search (BFS) and depth-first search (DFS) ● Indegree and outdegree ● directed and undirected graph	4	Easy	CO3		BL3
8	Write a program for Prim's Algorithm and explain the role of the visited array in avoiding cycles.	5	Moderate	CO1		BL3

9	The following BFS code contains logical errors. Identify and correct them:
	void BFS(int start)
	{
	queue<int> q;
	q.push(start);
	while(!q.empty())
	{
	int node = q.front();
	cout << node << " ";
	for(int i = 0; i < n; i++)
	{
	if(adj[node][i] == 1)
	{
	q.push(i);
	}
	}
	q.pop();
	}
	}
		5	Moderate	CO1		BL3

10	Explain different methods of representing a graph. Compare adjacency matrix and adjacency list with suitable examples.	5	Moderate	CO1		BL3

11	Write a code to represent a graph using an adjacency list. The code should: ● Accept number of vertices and edges ● Dynamically create the graph ● Display the adjacency list	5	Moderate	CO1		BL3
12	Write pseudo code of topological sorting and explain with an example.	5	Moderate	CO1		BL3
13	Starting from node M, determine a possible order in which the nodes are visited using the BFS algorithm. Also, illustrate the status of the queue after each step of traversal.  	5	Moderate	CO1	GATE	BL3
14	Explain how a stack is used to perform a depth first Traversal. Trace it for a given graph  	5	Moderate	CO1		BL3
15	Starting from node A, determine a possible order in which the nodes are visited using the DFS algorithm. Also, illustrate the status of the stack after each step of traversal.	5	Moderate	CO1		BL3

<table>
<tr><td></td><td> 
</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>16</td><td>A city planning department wants to connect multiple smart junctions using optical fiber cables such that the total installation cost is minimized. The possible connections between junctions along with their installation costs (in lakhs) are represented by the following weighted undirected graph.
 

Apply Prim’s Algorithm starting from vertex 6 to find minimum installation cost. Show the step-by-step selection of edges.</td><td>8</td><td>Moderate</td><td>CO1</td><td></td><td>BL3</td></tr>
<tr><td>14</td><td>Consider a weighted graph represented by the following adjacency matrix:
<table>
<tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr>
<tr><td>A</td><td>0</td><td>4</td><td>2</td><td>0</td><td>0</td></tr>
<tr><td>B</td><td>4</td><td>0</td><td>1</td><td>5</td><td>0</td></tr>
<tr><td>C</td><td>2</td><td>1</td><td>0</td><td>8</td><td>10</td></tr>
<tr><td>D</td><td>0</td><td>5</td><td>8</td><td>0</td><td>2</td></tr>
<tr><td>E</td><td>0</td><td>0</td><td>10</td><td>2</td><td>0</td></tr>
</table>
Apply Dijkstra’s Algorithm to find the shortest path from source vertex A to all other vertices.
Show the following at each iteration:</td><td>8</td><td>Moderate</td><td>CO1</td><td></td><td>BL3</td></tr>
</table>

<table>
<tr><td></td><td>

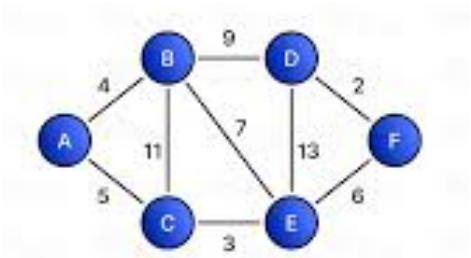
- Distance array
- Visited set
- Previous (parent) array

Also, write the final shortest path from **A to E**.

</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>18</td><td>Develop pseudo code of kruskal algorithm and explain with an example.</td><td>5</td><td>Moderate</td><td>CO1</td><td></td><td>BL3</td></tr>
<tr><td>19</td><td>Construct program to implement Prim's Algorithm using an adjacency matrix to find the Minimum Spanning Tree of a graph.</td><td>5</td><td>Moderate</td><td>CO1</td><td></td><td>BL3</td></tr>
<tr><td>20</td><td>Consider a graph represented by the following adjacency matrix:

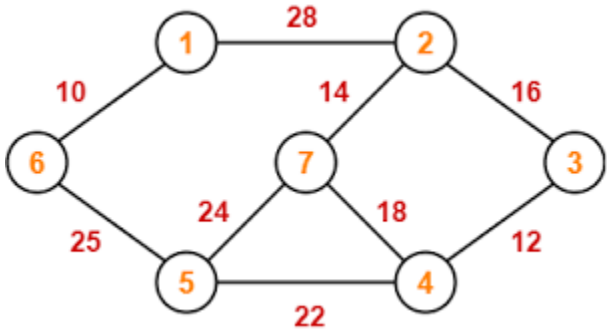
<table>
<tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr>
<tr><td>A</td><td>0</td><td>4</td><td>2</td><td>0</td><td>0</td></tr>
<tr><td>B</td><td>4</td><td>0</td><td>1</td><td>5</td><td>0</td></tr>
<tr><td>C</td><td>2</td><td>1</td><td>0</td><td>8</td><td>10</td></tr>
<tr><td>D</td><td>0</td><td>5</td><td>8</td><td>0</td><td>2</td></tr>
<tr><td>E</td><td>0</td><td>0</td><td>10</td><td>2</td><td>0</td></tr>
</table>

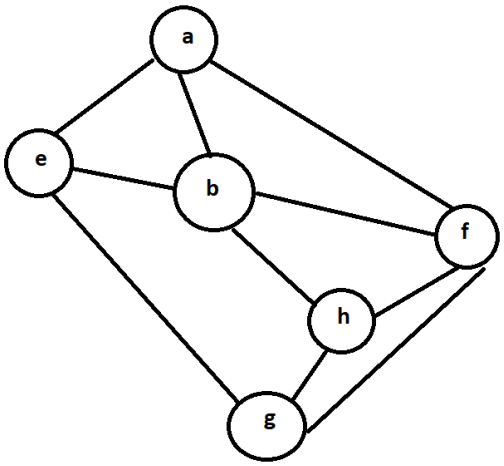
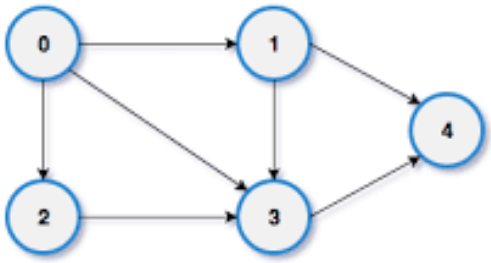
Apply Kruskal algorithm to find the minimum spanning tree and show the cost of MST.</td><td>7</td><td>Moderate</td><td>CO1</td><td></td><td>BL3</td></tr>
<tr><td>21</td><td>A company wants to connect its branch offices with minimum networking cost.



Using Kruskal Algorithm, construct the MST starting from vertex **A**. List the edges in the order they are selected and compute the final cost.</td><td>6</td><td>Moderate</td><td>CO1</td><td></td><td>BL3</td></tr>
<tr><td>22</td><td>A delivery company wants to find the shortest route for parcel delivery between different warehouses. The distances between warehouses (in km) are given below:

- A → B = 6
- A → D = 1</td><td>7</td><td>Moderate</td><td>CO1</td><td></td><td>BL3</td></tr>
</table>

	• D → B = 2 • B → E = 2 • D → E = 1 • E → C = 5 • B → C = 5 Apply Dijkstra's Algorithm to find the shortest path from warehouse **A to C**. Suggest the optimal route and total distance for delivery.					
23	A university is scheduling courses based on prerequisites as follows: • Course A must be completed before B and C • Course B must be completed before D • Course C must be completed before D and E • Course D must be completed before F Represent the above scenario as a Directed Graph. Apply Topological Sorting step by step to determine a valid order of course completion.	6	Moderate	CO1		BL3
24	Construct function to implement Kruskal Algorithm using an adjacency list to find the Minimum Spanning Tree of a graph.	5	Moderate	CO1		BL3
25	Construct function to implement dijikstras Algorithm using an adjacency matrix to find the shortest path. Illustrate step by step execution with examples.	5	Moderate	CO1		BL3
26	Construct function to implement Kruskal Algorithm using an adjacency matrix to find the Minimum Spanning Tree of a graph	5	Moderate	CO1		BL3
27	A company wants to connect its branch offices with **minimum networking cost**.  Using Prim's Algorithm, construct the MST starting from vertex **A**. List the edges in the order they are selected and compute the final cost.	7	Moderate	CO1		BL3

28	Consider the following graph  Among the following sequences: (I) a b e g h f (II) a b f e h g (III) a b f h g e (IV) a f g h b e Which are the depth-first traversals of the above graph?	5	Difficult	CO1	GATE	BL3
29	Construct function to display indegree and out degree of node from the given graph  	5	Difficult	CO1		BL3
30	Consider a directed graph with vertices representing tasks and edges representing dependencies. Explain how Topological Sorting can be used to detect cycles in a graph. Suggest a real-life situation where cycle detection is critical	5	Difficult	CO1		BL3
31	Write a program to detect whether a graph is connected or not using BFS. Explain the logic and demonstrate with an example.	5	Difficult	CO1		BL3
32	For a large-scale graph (e.g., social network), justify why adjacency list is preferred over adjacency matrix. Discuss in terms of: ● Memory usage	5	Difficult	CO1		BL3

	• Traversal efficiency • Scalability					

Unit 4

Indexing and Multiway Trees: Indexing: Introduction, Indexing techniques; B-Tree, B+ Tree; Red Black Tree; AA Tree; Splay Tree;

S No	Questions	Marks	Difficulty level	CO Mapping	GATE	BL
1	Discuss advantages and disadvantages of Splay Tree.	4	Easy	CO4		BL2
2	Construct a B+ Tree for given data and explain its structure.	5	Easy	CO4		BL2
3	Discuss indexing techniques in detail with diagrams and examples.	4	Easy	CO4		BL2
4	State what is multilevel indexing. Discuss its advantages over single-level indexing.	5	Easy	CO4		BL2
5	Discuss all rotation cases in Splay Tree with diagrams.	4	Easy	CO4		BL2
6	Discuss the limitations of single-level indexing and how multilevel indexing overcomes them.	5	Easy	CO4		BL2
7	How does indexing affect storage space and memory usage?	3	Easy	CO4		BL2
8	State four differences betrween Red-Black Tree and Splay Tree .	4	Easy	CO4		BL2
9	State four differences between AA Tree and Red-Black Tree.	4	Easy	CO4		BL2
10	Discuss Components and features of B+ Tree.	4	Moderate	CO4		BL2
11	Insert the following elements into a Red-Black Tree and show all steps: 10, 20, 30, 15, 25	5	Moderate	CO4		BL2

12	Illustrate deletion in Red-Black Tree with example.	6	Moderate	CO4		BL2
13	Construct a B-Tree of order 4 by inserting the following keys: 10, 20, 5, 6, 12	6	Moderate	CO4		BL2
14	Perform splaying operations for the sequence: 10, 20, 30, 40, 50	5	Moderate	CO4		BL2
15	Write an a function for B-Tree insertion	6	Moderate	CO4		BL2
16	Write an a function for B-Tree deletion	6	Moderate	CO4		BL2
17	Demonstrate skew and split operations of AA Tree with any example.	5	Moderate	CO4		BL2
18	Compare different indexing techniques.	6	Moderate	CO4		BL2
19	Discuss clustered vs non-clustered indexing with suitable examples.	6	Moderate	CO4		BL2
20	Construct a B-Tree of order 5 by inserting: 15, 25, 5, 10, 20 Show all intermediate steps.	6	Difficult	CO4		BL2
21	Insert the given sequence into a Red-Black Tree and show rotations and recoloring: 7, 3, 18, 10, 22 Show all intermediate steps.	6	Difficult	CO4		BL2
22	Construct an AA Tree for a given sequence and show skew and split operations. **10, 20, 30, 15, 5, 25**	6	Difficult	CO4		BL2
23	Write a function for Red-Black Tree insertion.	6	Difficult	CO4		BL2
24	Start with an empty RBT and insert the sequence 8,18,5,15, 17,25,40. Label the color of each node and show the intermediate steps	6	Difficult	CO4		BL2